

James Yihan Zhang

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EDUCATION

Yale University

Visiting Graduate Student: Financial Economics I, Econometrics I (First-Year PhD Courses)

Sep 2025 – Present

Grades: Honors

University of Maryland, College Park

B.S. Computer Science & Mathematics, Minor Computational Finance

Aug 2022 – May 2025

GPA: 3.9/4.0

EXPERIENCE

Yale University

*Pre-Doctoral Research Fellow, School of Management
with Bryan Kelly, Head of Machine Learning at AQR*

New Haven, Connecticut

Aug 2025 – Present

- Building a large language model-based annotation pipeline that reads Dow Jones Newswires, produces structured belief annotations, and assigns each story to the appropriate global futures markets for empirical finance applications.
- Developing a dynamic macroeconomic knowledge graph built from financial news and researching graph algorithms to investigate Shiller's famous "excess volatility" puzzle, linking the evolving flow of macro information to asset prices.
- Publishing "Global Macroeconomic Data" documentation (the macro analogue to Jensen, Kelly, and Pedersen 2023), covering returns of 92 futures contracts using data from LSEG Tick History, LSEG Datastream, and Compustat.

Independent Trading – Kalshi

Proprietary Prediction Markets Trader

New Haven, Connecticut

Jan 2026 – Present

- Generated a 19% year-to-date net return on \$14,000 of personal capital with a Sharpe ratio of 2.2.
- Deploying automated market making systems in Rust on an AWS Lightsail instance trading across WTI crude oil, Bitcoin, and other daily markets with real-time position tracking and risk controls.
- Using live CME Globex MDP 3.0 options data from Databento to implement pricing, volatility modeling, and delta hedging.
- Building a sportsbook that quotes same-game parlays on Kalshi, pricing correlated legs off FanDuel's parlay odds and quoting lines into live request-for-quote auctions to capture the vig across NBA, WNBA, MLB, and World Cup matches.

Amazon

Software Engineering Intern

Seattle, Washington

Jun 2025 – Aug 2025

- Spearheaded a real-time performance analytics project in Java, presenting key metrics to over 2 million active Amazon Sellers; led a database migration from Elasticsearch to AWS Timestream, projected to cut costs by 10% and user latency by 5%.
- Received a full-time return offer for the Amazon Video Shopping Experiences team.

University of Maryland, College Park

*Research Assistant, Department of Finance
with Serhiy Kozak*

College Park, Maryland

Jan 2024 – May 2025

- Designed a class of latent, three-dimensional factor models ("Tensor Factor Models") motivated by tensor decompositions, capturing drivers of asset returns across time, lags, and cross-sectional dimensions; implemented the model in Python using a dataset of 96 monthly factor portfolios.
- Wrote GPU-accelerated Orthogonalized Alternating Least Squares and PARAFAC algorithms in Jax, enabling scalable factorization of high-dimensional firm characteristics data.
- Implemented a kernel-methods model of the cross-section of returns, writing a custom Gaussian-kernel operation in C++/CUDA and registering it as a native, differentiable Jax primitive (JIT, vmap, autodiff) for GPU-scale estimation.

Capital One

Software Engineering Intern

McLean, Virginia

Jun 2024 – Aug 2024

- Built a Slackbot using FastAPI for the Data Remediation team, supporting Capital One's internal workflows and enabling faster resolution tracking on data lake platforms; improved query speeds by 15% and cut costs by 5%.

Northrop Grumman

Software Engineering Intern

Boulder, Colorado

Jun 2023 – Aug 2023

- Developed a time-varying atmospheric attenuation model with Chebyshev polynomials & approximation theory in PyTorch.
- Created a platform to generate atmospheric attenuation reference tables using Pandas and the Modtran C++ API.

PROFILE

Technical Skills: Python, Rust, Java, C++, CUDA, Polars, Jax, PyTorch, Huggingface, vLLM, NumPy, Pandas, LaTeX

Teaching Experience: Teaching Assistant for BUFN402 (Portfolio Management)

Professional Development: Yale School of Management Weekly Seminars in Finance, Yale Tobin Center Weekly Pre-Doctoral Seminars in Economics, Yale Empirical Asset Pricing Reading Group

Research Interests: Financial Machine Learning, Global Macro, Commodity Markets, Prediction Markets, Large Language Models, Market Microstructure, Random Matrix Theory